

the team has now planned to look for community-based alternatives once the product and its technology stack mature. “We began with proprietary solutions so that someone could handle the complexity and scale in the beginning. For instance, using AWS (Amazon Web Services) Kinesis, instead of Apache Kafka, provides multiple shards and easy scalability,” the founder mentions.

Being a great fan of open source technologies since the time he did his master’s in computer science at the Indian Institute of Science in Bengaluru, Trivedi wants to rely entirely on open source. PushEngage already uses the Nginx server to scale connections for sending push notifications to Web browsers, and there are plans to switch to Apache Kafka from the proprietary Kinesis solution — to use a server architecture solely based on open source technologies.

The process of enabling automation

PushEngage uses open APIs provided by Web browsers such as Chrome, Firefox and Safari that adhere to the W3C-Push standards. In addition to leveraging the available APIs, the company uses its internally built libraries that have Node.js as a programming language and MongoDB as a data store.

“The open source technologies behind PushEngage help us become more competitive as these solutions are tested thoroughly for scale, security and bugs. The deployment of open source also reduces our maintenance tasks and offers us the freedom to move across any cloud provider without a vendor lock-in,” Trivedi says.

Challenges of mass adoption

Even though open source has made PushEngage capable of delivering push notifications on a massive scale, catering to thousands of clients was not easy for the company in the early stages. “Push notifications send and receive messages in bursts, and hence the peak capacities are very high but of a short duration. Building a scalable solution around these issues requires well thought out architecture and solid engineering,” explains Trivedi.



The team at PushEngage that handles millions of notifications each day

The first version of the automated model was a minimally viable product. However, subsequent versions enabled the engineers to incorporate a highly scalable and efficient combination that helps to send and receive messages on a large scale.

Apart from building just a scalable solution, the PushEngage team was required to set up a datastore that could generate queries on several attributes in real-time and push messages at a fast pace. There was also a need for datastores that had high read characteristics.



PushEngage enables automation on the Web

PushEngage moved to a microservice-powered, asynchronous message-based architecture to overcome the primary challenges. “The deployment of appropriate architecture provided us with the desired burst-mode scalability as well as fast enough results for our customers,” says Trivedi.

The role of the community

To build a secure and advanced logging system, as well as testing frameworks for automating notifications, PushEngage took help from the open source community, accessing libraries on GitHub and SourceForge. “Online listings of open source developments are quite handy in finding good libraries, so we don’t have to

re-invent the wheel,” says Trivedi.

PushEngage is also committed to giving back to the community and developers worldwide. The company has already provided a push notification API free to developers. This allows any developer to integrate all the rich features of customer segmentation, automation, scheduling and triggering notifications, as well as geotargeting.

Open source adoption cuts costs

According to Trivedi, deploying open source helped his company to reduce the cost of the project by 25 per cent. “Open source adoption enabled PushEngage to reduce not just initial operational costs but also our ongoing cost of product development,” he notes.

Future-proof

Moreover, open source is making the automation process through PushEngage future-proof. “We have a microservice-based architecture built for horizontal scaling. We also use queue-based message passing, as well as datastores that are strong in both read and write characteristics and have a scalable database service. All this ensures our scaling needs are well met,” says Trivedi.

The company also uses relational and non-relational databases, depending on the need of the architecture. All this makes the push notification technology capable of expanding, with the addition of new components and features in the future. **END**

By: Jagmeet Singh

The author is an assistant editor at EFY.